Rohan Puntambekar

Portland, ME | Email | linkedin.com/in/rohan | GitHub

Education

The Roux Institute at Northeastern University — Portland, ME, USA
Master of Science in Artificial Intelligence — *09/2024 – Expected: 08/2026*

Institute of Technology, Nirma University — Ahmedabad, Gujarat, India
Bachelor of Technology in Mechanical Engineering — *10/2020 – 06/2024*

Research Experience

Graduate Researcher — 09/2025 – Present
The Roux Institute, Northeastern University | Advisor: Dr. Michael Wan | Master's Thesis — *Portland, Maine, USA*

- Investigating regression-aware feature transformation for domain adaptation in computer vision tasks, focusing on representation learning that preserves geometric properties across domains.
- Benchmarking existing domain adaptation algorithms on vision datasets while analyzing covariate shift effects.
- Developing novel coefficient of variation-based unsupervised domain adaptation approach for regression tasks.
- Designing experiments to compare model performance under varying domain shifts and feature transformation.

Undergraduate Researcher — 12/2022 – 06/2024
Nirma University | Advisors: Dr. Ankit Thakkar, Dr. Dhaval Patel — *Ahmedabad, Gujarat, India*

- Conducted extensive literature survey of machine learning and deep learning methods for bearing fault diagnosis, focusing on advanced architectures such as GNNs, DRL, attention mechanisms, and transformer models.
- Performed critical analysis of benchmark datasets and SOTA methods, highlighting limitations such as dataset bias, data imbalance, interpretability and real-time adoption failure.
- Designed and validated experimental frameworks comparing various deep learning architectures and feature selection methods, identifying recurrent networks and CNN-based extraction as optimal.
- Developed a novel automated resonant frequency band identification using Z-score and median-based measures.
- Implemented statistical measure–based feature discretization for efficient deep learning model training.
- Developed signal fusion–based transfer learning with MCNN-LSTM framework for cross-domain fault detection.
- Research resulted in five journal papers: one published, three under review, and one under revision.

Professional Experience

Machine Learning Engineer — 07/2025 – 12/2025
Apriqot Inc. — *Portland, Maine*

- Developed a statistical modeling framework combining a GLM with logit link and population characterization model to estimate household food insecurity at subcounty levels across Maine.
- Processed multi-scale survey data and applied Monte Carlo simulation for uncertainty quantification.
- Designed a novel hierarchical proportional model for diverse demographic subgroups, propagating estimates from PUMA-level data through geographic hierarchy using adapted iterative proportional fitting.
- Created choropleth visualizations highlighting spatial patterns and demographic disparities, supporting health improvement planning, program targeting, and resource allocation.

Product Design & Automation Intern — 06/2023 – 07/2023
Xylem Water Solutions India Pvt. Ltd. — *Vadodara, Gujarat*

- Contributed to the design, development, and drafting of 5+ mechanical components using 3-D modeling principles.
- Collaborated with a multidisciplinary team on research to digitize and automate product design processes.

Publications

- **Puntambekar, R.**, Vyas, P., Thakkar, A., & Patel, D. (2025). A survey of machine learning and deep learning methods for vibration-based bearing fault diagnosis: The need, challenges, and potential future research directions. *Neurocomputing*, 131628.
- **Puntambekar, R.**, Thakkar, A., & Patel, D. Deep learning models and feature selection techniques for bearing fault detection: A comparative study. *Computer and Electrical Engineering*. (Under review).
- **Puntambekar, R.**, Thakkar, A., & Patel, D. Feature discretization-based data processing for improving computational aspects in deep learning-based bearing fault detection. *Engineering with Computers*. (Under review).
- Patel, D., Thakkar, A., **Puntambekar, R.**, Vyas, P., Bhojawala, V., & Patel, K. Fusion of outlier detection and statistical measures to identify resonant frequency band for bearing fault detection. *Mechanical Systems and Signal Processing*. (Under review).
- Thakkar, A., **Puntambekar, R.**, & Patel, D. Signal fusion-based domain adaptation for bearing fault detection. (Under revision).

Presentations

Maine Public Health Association 2025 Annual Conference, *Augusta, Maine* October 2025
- Oral presentation on "Integrating Food Insecurity Data Systems to Support Public and Population Health Action"

UG SRC Research Symposium, *Ahmedabad, Gujarat* March 2024
- Poster presentation on "Improvement in Deep Learning–Based Bearing Fault Detection Using Feature Discretization."

Academic Projects

Retrieval-Augmented Generation for Query Answering on Construction Industry Standards 01/2025 – 04/2025
- Led the development of a RAG pipeline for construction standards, curating a 300 Q&A pair ground truth dataset and implementing semantic-aware document chunking for high-fidelity retrieval.
- Directed an ablation study across embeddings and generative models, identifying IBM Granite embeddings with Gemma3 1b (temperature 0.5) as an optimal combination.

Equity Assessment in post-HCT Survival Predictions 01/2025 – 04/2025
- Developed fairness-aware survival prediction models for allogeneic hematopoietic cell transplantation to mitigate racial and socioeconomic disparities in clinical outcome predictions.
- Applied mutual information and PCA for feature reduction and optimized survival models through cross-validation.

Deep Learning models for Intrusion Detection in Autonomous Vehicle Networks 09/2024 – 12/2024
- Evaluated deep learning models for CAN traffic dataset, testing multiple architectures and sampling techniques.
- Identified LSTM with Synthetic Minority Oversampling as optimal configuration.

Honors and Awards

2nd Place, Climate Resiliency Hackathon, *Northeastern University* October 2024
Alfond Scholar, $25000 merit scholarship, *The Roux Institute, Northeastern University* September 2024
Silver Medal, Academic Performance, Department of Mechanical Engineering, *Nirma University* August 2024
Certificate of Academic Excellence, Department of Mechanical Engineering, *Nirma University* August 2024
1st Prize, Poster Presentation (Track 5-Mechanical), UG SRC Research Symposium, *Nirma University* March 2024

Workshops & Certificates

Introduction to Deep Learning Boot camp, Alexander & Ava Amini, *Massachusetts Institute of Technology* January 2025
Foundational Certificate in Programming and Data Science, *IIT Madras* April 2022
IoT and Automation using NodeMCU Workshop, *Nirma University* March 2022
FC1x: Fat Chance: Probability from the Ground Up Certificate, *Harvard University* November 2020

Service & Outreach

Volunteer, Data for Social Good, *Northeastern University* 09/2024 – 12/2024
Representative Member for Institute of Technology, Women Development Cell, *Nirma University* 07/2023 – 12/2023
Creative Head, Storyteller's Club, *Nirma University* 11/2022 – 05/2023
Executive Committee Member, Anveshan TechFest 2022, *Nirma University* 02/2022 - 04/2022

Technical Skills

Programming Languages: Python, MATLAB, C/C++, Java
AI/ML Libraries: PyTorch, Keras, TensorFlow, scikit-learn, pandas, NumPy, Matplotlib, Seaborn, NLTK
Data Engineering & BI Tools: SQL, Tableau, Power BI, Google Cloud Platform, Microsoft Azure
Developer Tools/Platforms: Git, Docker, Bash/Shell Scripting